MCA SEM-II Syllabus										
Sr No	**Category**	**Course Code**	**Subject**	**L**	**T**	**P**	**CA**	**ESE**	**Total**	**Credit**
1	PCC	MC2T001	Data Base Management System	4	0	0	40	60	100	4
2	PCC	MC2T002	Artificial Intelligence	4	0	0	40	60	100	4
3	PCC	MC2T003	Data Science Using Python	3	0	0	40	60	100	3
4	PCC	MC2T004	Cyber Security & Cryptography	3	0	0	40	60	100	3
5	PEC	MC2E005	Department Elective-I	2	0	0	40	60	100	2
6	PCC	MC2L006	Data Base Management System LAB	0	0	4	60	40	100	2
7	PCC	MC2L007	Python LAB	0	0	4	60	40	100	2
8	PCC	MC2L008	PHP LAB	0	0	4	60	40	100	2
			Total	**16**	**0**	**12**	**380**	**420**	**800**	**22**

ELECTIVE-I Basket (MC2E005)
PHP
Computer Graphics
Research Methodology

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2T001	Database Management System	4	-	-	4

Marks Distribution			
CA	**MSE**	**ESE**	**Total**
40	-	60	100

Prerequisites for the course	
1	Basic understanding of computer systems and operating environments.

Prior Reading Material/useful links	
1	https://www.tutorialspoint.com/dbms/index.htm
2	https://www.w3schools.com/sql/sql_intro.asp
3	https://archive.nptel.ac.in/courses/106/105/106105175/

Course Objectives	
Sr No	**Statement**
1	Provide a strong foundation in database concepts, including the architecture, applications, and purposes of database systems.
2	Equip students with the skills to design and model databases using Entity-Relationship diagrams, understanding entities, attributes, relationships, and associated issues.
3	Introduce students to the relational model and normalization techniques, focusing on reducing redundancy and improving data integrity.
4	Develop a solid understanding of SQL and other relational query languages, enabling students to perform data manipulation, querying, and management effectively.
5	Familiarize students with the concepts of transaction management, including properties, concurrency control, and recovery techniques, ensuring the reliability and integrity of database operations.

Course Outcomes		
Sr No	**Code**	**CO statement**
1	CO1	Explain Core Concepts and Architectures of Database Systems
2	CO2	Design and Model Databases Using ER Diagrams
3	CO3	Apply SQL for Database Manipulation and Querying
4	CO4	Analyze and Normalize Database Schemas
5	CO5	Manage Transactions and Implement Concurrency Control

Unit No	Course Contents	Duration
I	Database Management System – Concepts and Architectures Database System Applications, Purpose of Database Systems, View of Data , Data Abstraction (Instances and Schemas), data Models , the ER Model , Relational Model, Database Languages (DDL , DML, DCL, and TCL). Data base design and ER diagrams, ER Model , Entities, Attributes, and Entity sets ,Relationships and Relationship sets , ER Design Issues	6 hours
II	Relational Query Languages Introduction to the Relational Model , Structure , Database Schema, Keys Schema Diagrams, Overview of the SQL Query Language – Basic Structure of SQL Queries, Set Operations, Aggregate Functions (GROUPBY – HAVING, Nested) Sub queries, joins , Triggers.	7 hours
III	Normalization Introduction, Non loss decomposition and functional dependencies, First, Second, and third normal forms – dependency preservation, Boyee/Codd normal form, Higher Normal Forms, Introduction, Multi-valued dependencies and Fourth normal form.	7 hours
IV	Transaction Concept Introduction What is a Transaction? , Transaction Properties , Transaction Management with SQL , The Transaction Log Concurrency Control , Concurrency control with Locking Methods , Types of Locks , Two-Phase Locking to Ensure Serializability , Deadlocks, , Concurrency Control Methods.	8 hours
V	File organization : File organization, various kinds of indexes. Query Processing , Measures of query cost , Selection operations, Join operations, join operations, set operation and aggregate operation	8 hours

Text Books				
Code	**Title of the Book**	**Author Name /Designation / Organization**	**Publisher**	**Edition/ Publication Year**
T1	Database Systems: The Complete Book	Hector Garcia-Molina, Jeffrey D. Ullman, and Jennifer Widom	Pearson	2021
T2	Database Management Systems	S.K. Singh	McGraw-Hill Education	2022

Reference Books				
Code	**Title of the Book**	**Author Name/Designation/ Organization**	**Publisher**	**Edition/ Publication Year**
R1	Database Management Systems	Raghu Ramakrishnan and Johannes Gehrke	McGraw-Hill Education	2020

R2	Fundamentals of Database Systems	Ramez Elmasri and Shamkant B. Navathe	Pearson	2021

Useful Links	
1	https://www.tutorialspoint.com/dbms/index.htm
2	https://www.w3schools.com/sql/sql_intro.asp

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2T002	Artificial Intelligence	4	-	-	4

Marks Distribution			
CA	**MSE**	**ESE**	**Total**
40	-	60	100

Prerequisites for the course	
1	Requires some programming experience and some mathematical fluency.
2	Need a very strong background in linear algebra

Prior Reading Material/useful links	
1	https://see.stanford.edu/Course/CS229
2	http://www.cognitivelearningbook.org/

Course Objectives	
Sr No	**Statement**
1	Provide a comprehensive introduction to Artificial Intelligence, focusing on its key concepts, techniques, and applications.
2	Develop students' problem-solving skills through the implementation of AI search strategies and algorithms.
3	Explore and understand game-playing algorithms and their application in AI.
4	Introduce knowledge representation, reasoning under uncertainty, and various learning methods used in AI.
5	Discuss the role of AI in cognitive robotics and the ethical considerations associated with AI technologies.

Course Outcomes		
Sr No	**Code**	**CO statement**
1	CO1	Demonstrate understanding of the fundamental concepts and features of Artificial Intelligence, including agents and their environments.
2	CO2	Apply various search strategies, including uninformed and heuristic searches, to solve AI problems effectively.
3	CO3	Analyze game-playing algorithms and implement solutions using minimax, alpha-beta pruning, and evaluation functions for optimal decision-making.
4	CO4	Explain and apply knowledge representation techniques, probabilistic reasoning, and learning algorithms to AI models and real-world problems.
5	CO5	Evaluate the ethical implications and risks associated with AI, particularly in the context of cognitive robotics and real-world case studies.

Unit No	Course Contents	Duration
I	Introduction to Artificial Intelligence, Features of AI, Agents and Environments, structure of agents, problem solving agents, problem formulation, AI techniques- search knowledge.	6 hours
II	Searching- Searching for solutions, uniformed search strategies – Breadth first search, depth first Search. Search with partial information (Heuristic search) Hill climbing, A* ,AO* Algorithms, Problem reduction, Game Playing-Adversial search, Games, mini-max algorithm, optimal decisions in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions.	7 hours
III	Knowledge Representation; Learning, Uncertainty, probabilistic reasoning-Bayesian Network, probabilistic reasoning over time-Inference in temporal Model, Hidden Markov models- Kalman filters, Dynamic Bayesian Network, speech recognition	7 hours
IV	Learning: Concept of learning, learning automation, genetic algorithm, learning by inductions, neural nets. Programming Language: Introduction to programming Language. Handling Uncertainties: Non-monotonic reasoning, Probabilistic reasoning, use of certainty factors, Fuzzy logic	8 hours
V	AI in Cognitive Robotics: Robotic perception, localization, mapping configuring space, planning uncertain movements, dynamics and control of movement, Ethics and risks of artificial intelligence in robotics. Case study of AI in robotics	8 hours

Text Books				
Code	**Title of the Book**	**Author Name /Designation / Organization**	**Publisher**	**Edition/ Publication Year**
T1	Introduction to Artificial Intelligence	Wolfgang Ertel	Springer	2023
T2	Artificial Intelligence: Foundations of Computational Agents	David L. Poole and Alan K. Mackworth	Cambridge University Press	2023

Reference Books				
Code	**Title of the Book**	**Author Name/Designation/ Organization**	**Publisher**	**Edition/ Publication Year**
R1	Artificial Intelligence: A Modern Approach	Stuart Russell and Peter Norvig	Pearson Education	2023
R2	Deep Learning	Ian Goodfellow, Yoshua Bengio, and Aaron Courville	MIT Press	2022

Useful Links	
1	https://nptel.ac.in/courses/106105077/
2	https://www.coursera.org/courses?query=robotics%20engineer
3	http://www.coursera.org/specializations/artificial-intelligence-overview

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2T003	Data Science using Python	3	-	-	3

Marks Distribution			
CA	**MSE**	**ESE**	**Total**
40	-	60	100

Prerequisites for the course	
1	Intermediate to Advanced knowledge in statistics is desired for data science work.

Prior Reading Material/useful links	
1	https://elearn.nptel.ac.in/shop/iit-workshops/completed/data-analysis-using-statistical-learning-techniques/
2	http://www.youtube.com/watch?v=I10q6fjPxJ0

Course Objectives	
Sr No	**Statement**
1	Provide an introduction to data science and familiarize students with the data science process and key terminologies.
2	Equip students with Python programming skills necessary for data science applications, including scripting and working in virtual environments.
3	Teach students data cleaning techniques using Python, enabling them to handle and preprocess data effectively.
4	Introduce Numpy and Pandas libraries for data processing, highlighting their importance and functionality in managing data.
5	Develop students' ability to visualize data using the Matplotlib library, teaching them how to present data insights through various types of plots.

Course Outcomes		
Sr No	**Code**	**CO statement**
1	CO1	Understand the fundamentals of data science, including the types of data, and the roles and responsibilities of a data scientist.
2	CO2	Demonstrate proficiency in Python programming for data science, including setting up a virtual environment and scripting fundamentals.
3	CO3	Apply data cleaning techniques using Python, including data inspection, handling missing data, outliers, and data transformation.
4	CO4	Utilize Numpy and Pandas libraries for efficient data processing and perform basic operations on arrays and dataframes.
5	CO5	Create data visualizations using Matplotlib, including line charts, bar charts, histograms, scatter plots, and pie charts to present data insights effectively.

Unit No	Course Contents	Duration
I	**Introduction to Data science:** Introduction to Data Science, Process of Data Science, types of Data, Big Data Vs Small Data, Basic terminologies of DS, Structured vs. Unstructured, Data Roles and Responsibility of Data scientist.	6 hours
II	**Introduction to python:** Need of python in data science, Python for data Scientist, Virtual environment for python, variables, rules for python variables, Fundamentals of Python and working with script.	7 hours
III	**Data cleaning using python:** Overview of data cleaning, Steps Involved in Data Cleaning: Data inspection and exploration, Handling missing data, Handling outliers, Data transformation, Advantages & disadvantages of Data Cleaning.	7 hours
IV	**Data Processing with Python:** Need of Numpy in data Processing, Powerful properties of numpy , Types of arrays, basic Operations using Numpy, Need of Pandas in data processing.	8 hours
V	**Data Visualization:** Using Matplotlib Library –Introduction to matplotlib plots, different types of matplotlib plots: Line chart , Bar chart, Histogram, scatter plot, Pie chart.	8 hours

Text Books				
Code	**Title of the Book**	**Author Name /Designation / Organization**	**Publisher**	**Edition/ Publication Year**
T1	Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython	Wes McKinney	O'Reilly Media	2nd Edition, 2017
T2	Think Python: How to Think Like a Computer Scientist	Allen B. Downey	O'Reilly Media	2nd Edition, 2015

Reference Books				
Code	**Title of the Book**	**Author Name/Designation/ Organization**	**Publisher**	**Edition/ Publication Year**
R1	Data Science from Scratch: First Principles with Python	Joel Grus	O'Reilly Media	2nd Edition, 2019
R2	Python for Data Science Handbook: Essential Tools for Working with Data	Jake VanderPlas	O'Reilly Media	2016

Useful Links	
1	http://www.coursera.org/learn/statistical-analysis-hypothesis-testing-sas
2	http://www.coursera.org/projects/statistical-analysis-using-python-numpy

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2T004	Cyber Security & Cryptography	4	-	-	3

Marks Distribution			
CA	**MSE**	**ESE**	**Total**
40	-	60	100

Prerequisites for the course	
1	Being aware of the network architecture and protocol

Prior Reading Material/useful links	
1	http://nptel.ac.in/courses/106105031/
2	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-033-com

Course Objectives	
Sr No	**Statement**
1	Provide an overview of cyber security, including key concepts, threats, and the importance of cyber security policies and regulations.
2	Introduce cryptographic techniques, including block ciphers, public key cryptosystems, and digital signatures, to enhance data security.
3	Explore key management and distribution methods, along with understanding vulnerabilities in complex network architectures.
4	Investigate cyber security safeguards, including access control, authentication, and ethical hacking practices.
5	Familiarize students with securing web applications, services, and understanding cyber forensic techniques.

Course Outcomes		
Sr No	**Code**	**CO statement**
1	CO1	Understand the foundational concepts of cyber security, including cyber threats, governance, and the need for comprehensive security policies.
2	CO2	Analyze and apply various cryptographic principles and techniques, including symmetric and asymmetric key cryptography.
3	CO3	Evaluate public key cryptosystems, cryptographic hash functions, and digital signatures to ensure data integrity and authenticity.
4	CO4	Examine key management and distribution methods, including Kerberos, X.509, and other authentication services, while identifying common cyber security vulnerabilities.
5	CO5	Implement security measures for web applications, servers, and explore cyber forensic techniques, intrusion detection, and prevention mechanisms.

Unit No	Course Contents	Duration
I	**Introduction to Cyber Security:** Overview of cyber security, Internet Governance-Challenges and constraints, Cyber threats:- Cyber Warfare-Cyber Crime-Cyber terrorism ,Cyber Espionage, Need for comprehensive cyber security policy, need for nodal authority, Cyber security regulations, Roles of international law.	6 hours
II	**Cryptography and Block Ciphers principles:** Introduction, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security, Substitution and Transposition techniques, Symmetric and Asymmetric key cryptography, Steganography, Cryptographic independent dimensions. Cryptanalytic attack and brute force attack, Symmetric key Ciphers: Block Cipher principles, DES.	7 hours
III	**Public Key Cryptosystems and Authentication Requirements:** Principles of public key cryptosystems, RSA algorithm, Diffie-Hellman Key Exchange, introductory idea of Elliptic curve cryptography. Cryptographic Hash Functions: Message Authentication, Secure Hash Algorithm (SHA-512) Message authentication codes: Authentication requirements, Digital signature.	7 hours
IV	**Key Management, Distribution and Cyber Security Vulnerabilities**: Distribution of Public Keys, Kerberos, X.509 Authentication Service, PGP, SSL, IPSEC. Cyber Security Vulnerabilities-Overview, vulnerabilities in software, System administration, Complex Network Architectures, Weak Authentication, Poor Cyber Security Awareness. Cyber Security Safeguards- Overview, Access control, Audit, Authentication, Ethical Hacking	8 hours
V	**Securing Web Application Services, Servers and cyber forensic:** Basic security for HTTP Applications and Services, Basic Security for SOAP Services, Identity Management and Web Services, Security Considerations, Challenges, Intrusion detection and Prevention Techniques, System Integrity Validation, Honey pots, password management. Introduction to Cyber Forensics.	8 hours

Text Books				
Code	**Title of the Book**	**Author Name /Designation / Organization**	**Publisher**	**Edition/ Publication Year**
T1	Cyber security for Beginners	Raef Meeuwisse	2022	Cyber Simplicity Ltd
T2	Network Security Essentials: Applications and Standards	William Stallings	2023	Pearson

Reference Books				
Code	**Title of the Book**	**Author Name/Designation/ Organization**	**Publisher**	**Edition/ Publication Year**
R1	Cryptography and Network Security: Principles and Practice	William Stallings	Pearson	2023
R2	Principles of Cybersecurity and Cyber Ethics: Digital Citizenship in a Global Society	Linda Wilbanks	Routledge	2023

Useful Links	
1	http://nptel.ac.in/courses/106105031
2	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-033-comput

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2E005	PHP	2	-	-	2

Marks Distribution			
CA	**MSE**	**ESE**	**Total**
40	-	60	100

Prerequisites for the course	
1	Basic understanding of HTML and CSS.
2	Understanding of web browsers and how websites work.

Prior Reading Material/useful links	
1	PHP Manual - Official Documentation
2	W3Schools PHP Tutorial
3	MySQL Documentation

Course Objectives	
Sr No	**Statement**
1	To introduce the fundamentals of PHP and its role in web development.
2	To learn how to create dynamic web applications using PHP.
3	To understand the integration of PHP with databases using MySQL.
4	To develop skills in handling web forms, file management, and session management in PHP.
5	To explore advanced PHP features and apply them in developing a complete web application.

Course Outcomes		
Sr No	**Code**	**CO statement**
1	CO1	Understand and apply basic PHP syntax, variables, data types, and control structures.
2	CO2	Develop dynamic web pages by integrating PHP with HTML and managing form data.
3	CO3	Implement PHP functions, arrays, and string manipulations in web applications.
4	CO4	Integrate PHP with MySQL to perform CRUD operations and manage databases securely.
5	CO5	Develop and deploy a full-fledged web application using advanced PHP techniques and frameworks.

Unit No	Course Contents	Duration
I	**Introduction to PHP and Setup** - Overview of PHP: Features, Benefits, Applications - PHP Installation: XAMPP, WAMP Setup - Basic Syntax: Variables, Data Types, Constants - Control Structures: If-Else, Switch, Loops - PHP in HTML: Embedding PHP within HTML Pages	7 hours
II	**PHP Functions and Working with Data** - Functions: Creation, Usage, Built-in Functions - Arrays: Types, Array Functions, Sorting, Filtering - String Manipulation: Functions, Regular Expressions - Working with Dates and Times	7 hours
III	**Web Forms, File Handling, and State Management** - Form Handling: Methods, Validation, Security - File System Functions: Reading, Writing, Uploading Files - Session and Cookie Handling: Managing User Data - Security Measures: Input Filtering, Output Escaping	7 hours
IV	**Database Connectivity and Data Management** - Introduction to Databases: SQL Basics, MySQL - PHP and MySQL: Connecting, CRUD Operations - Advanced Database Concepts: Joins, Indexes, Transactions - Database Security: Preventing SQL Injection, Best Practices	8 hours
V	**Advanced Topics and Project Development** - Object-Oriented PHP: Classes, Objects, OOP Principles - Advanced PHP Features: Namespaces, Traits, Anonymous Classes - Building RESTful APIs with PHP - Introduction to PHP MVC Frameworks: Laravel Basics - Project Work: Develop a Complete Web Application	8 hours

Text Books				
Code	**Title of the Book**	**Author Name /Designation / Organization**	**Publisher**	**Edition/ Publication Year**
T1	PHP & MySQL: Novice to Ninja	Kevin Yank	SitePoint	6th Edition, 2017
T2	PHP for the Web: Visual QuickStart Guide	Larry Ullman	Peachpit Press	5th Edition, 2016

Reference Books				
Code	**Title of the Book**	**Author Name/Designation/ Organization**	**Publisher**	**Edition/ Publication Year**
R1	Learning PHP, MySQL & JavaScript	Robin Nixon	O'Reilly Media	5th Edition, 2018
R2	Modern PHP: New Features and Good Practices	Josh Lockhart	O'Reilly Media	1st Edition, 2015

Useful Links	
1	https://phptherightway.com/
2	https://www.w3schools.com/php
3	https://dev.mysql.com/doc/refman/8.0/en/

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2E005	Research Methodology	3	-	-	2

Marks Distribution			
CA	**MSE**	**ESE**	**Total**
40	-	60	100

Prerequisites for the course	
1	Familiarity with academic writing conventions and structures.
2	Basic statistical concepts and their applications in research.

Prior Reading Material/useful links	
1	https://onlinecourses.nptel.ac.in/noc23_ge36/preview
2	https://onlinecourses.nptel.ac.in/noc22_ge08/preview
3	https://www.scribbr.com/dissertation/methodology/

Course Objectives	
Sr No	**Statement**
1	Introduce students to the basic concepts and objectives of research and to familiarize them with the research process.
2	Enable students to distinguish between different types of research and to identify suitable research methods for various research questions.
3	Provide knowledge on sampling techniques, research design, and their applications in conducting effective and unbiased research.
4	Equip students with skills in data collection, analysis, and the use of computer applications for data processing in research.
5	Develop competencies in presenting research findings, adhering to ethical standards, and mastering various citation styles and report writing formats.

Course Outcomes		
Sr No	**Code**	**CO statement**
1	CO1	Understand the fundamentals of research, including the meaning, objectives, process, and differentiation between methods and methodology.
2	CO2	Distinguish between various types of research such as pure, applied, exploratory, descriptive, diagnostic, quantitative, and qualitative research.
3	CO3	Apply knowledge of sampling techniques and research design to develop appropriate research frameworks and reduce sampling errors.
4	CO4	Execute methods for data collection, analysis, and presentation, including primary and secondary data sources, data classification, and visualization techniques.
5	CO5	Demonstrate proficiency in research presentation, including citation styles, ethical considerations, plagiarism avoidance, and report writing.

Unit No	Course Contents	Duration
I	Fundamentals of research; Meaning, Objectives, Research process, Methods and Methodology, Criteria of good research, Reviewof literatures: Primary source, Secondary source, Identifying gap areas from literature review, Searchinge- resources, using search engines, Searching data base.	6 hours
II	Types of Research; Pure research, applied research, Exploratory Research, Descriptive research, Diagnostic research, Quantitative and Qualitative research.	7 hours
III	Research Sampling and Design: Sampling of data: Concept of sampling, Probability sampling techniques, Non probability sampling techniques, Sampling error, Research Design: Meaning, Need, Types of research design-Exploratory Research Design, components of research design and features of good research design	7 hours
IV	Methods, Collection and Analysis of Data: Types of data, Methods of data collection- Interview Method, Mailing Method, Observation Method, Survey Method etc.; Primary and secondary sources of data, Sampling- meaning and methods, Classification and Tabulation, Graphical presentation, Application of computer in research data analysis.	8 hours
V	Presentation of Research: Citation Styles- APA, MLA etc., Research ethics and Plagiarism, Indexing of journal and research output, Report writing steps in report writing, layout of report writing, reference and bibliography.	8 hours

Text Books				
Code	**Title of the Book**	**Author Name /Designation / Organization**	**Publisher**	**Edition/ Publication Year**
T1	Business Research Methods	Donald R. Cooper, Pamela S. Schindler	McGraw-Hill Education	2023
T2	Research Design: Qualitative, Quantitative, and Mixed Methods Approaches	John W. Creswell	SAGE Publications	2023

Reference Books				
Code	**Title of the Book**	**Author Name/Designation/ Organization**	**Publisher**	**Edition/ Publication Year**
R1	Research Methodology: A	Ranjit Kumar	SAGE Publications	2021

	Step-by-Step Guide for Beginners			
R2	Research Methods in Education	Louis Cohen, Lawrence Manion, and Keith Morrison	Routledge	2022

Useful Links	
1	https://www.scribbr.com/dissertation/methodology/
2	https://onlinecourses.nptel.ac.in/noc23_ge36/preview

Semester	Course Code	Name of the course	L	T	P	Credits
II	MC2E005	Computer Graphics	3	-	-	2

Marks Distribution			
CA	MSE	ESE	Total
40	-	60	100

Prerequisites for the course	
1	Basic knowledge of programming in C/C++.
2	Familiarity with fundamental mathematics concepts including linear algebra and geometry.

Prior Reading Material/useful links	
1	http://www.tutorialspoint.com/computer_graphics/computer_graphics_quick_guide.htm
2	http://www.javatpoint.com/computer-graphics-tutorial
3	http://www.coursera.org/articles/computer-graphics

Course Objectives	
Sr No	Statement
1	Introduce students to the foundational concepts of computer graphics, including hardware and software components.
2	Equip students with the skills to implement basic raster graphics algorithms for drawing and filling geometric shapes.
3	Familiarize students with the OpenGL programming environment for creating and manipulating 2D and 3D graphics.
4	Teach students how to perform transformations and viewing operations on graphical objects, both in 2D and 3D spaces.
5	Provide students with an understanding of advanced graphics topics such as clipping, hidden surface removal, and animation techniques to achieve realism in computer graphics.

Course Outcomes		
Sr No	Code	CO statement
1	CO1	Understand the fundamental concepts of computer graphics, including the graphics pipeline, graphics APIs, and display technologies.
2	CO2	Develop and implement algorithms for drawing 2D primitives, polygon filling, and applying anti-aliasing techniques.
3	CO3	Utilize OpenGL for graphics programming, including creating animations and implementing basic 3D viewing and transformation operations.
4	CO4	Apply 2D and 3D geometric transformations, including affine transformations and coordinate system conversions, to manipulate graphical objects.

5	CO5	Demonstrate knowledge of viewing transformations, 3D projections, hidden surface removal techniques, and animation sequences for realistic graphics rendering.

Unit No	Course Contents	Duration
I	Introduction to Computer Graphics: Overview of Computer Graphics, Computer Graphics Application and Software, Graphics Areas, Graphics Pipeline, Graphics API's, Numerical issues, Efficiency Display and Hardcopy Technologies, Display Technologies – Raster scan Display System	6 hours
II	Basic Raster Graphics: Algorithms for Drawing 2D primitives, aliasing and ant aliasing, Polygon filling methods: Scan Conversion Algorithms: Simple Ordered edge list, Edge Fill, Fence fill and Edge Flag Algorithm, Seed fill Algorithms: Simple and Scan Line Seed Fill Algorithm, Halftoning techniques.	7 hours
III	Graphics Programming using OPENGL: Why OpenGL, Features in OpenGL, OpenGL operations, Abstractions in OpenGL – GL, GLU & GLUT, 3D viewing pipeline, viewing matrix specifications, a few examples and demos of OpenGL programs, Animations in open GL.	7 hours
IV	2-Dgeometric transformations: Basic transformations, matrix representations, composite transformations, other transformations, transformations between coordinate systems, affine transformations, transformation functions, Raster methods for transformations. TwoDimensional viewing : viewing coordinates, Window-to viewport coordinate transformation, viewing functions, clipping : point, line, polygon, curve, text, exterior.	8 hours
V	Normalized Device Coordinates and Viewing Transformations:3D System Basics and 3D Transformations, 3D graphics projections, parallel, perspective, viewing transformations.3D graphics hidden surfaces and line removal, painter's algorithm, Z -buffers, Warnock's algorithm. Animations & Realism 10 Animation Graphics: Design of Animation sequences – animation function – raster animation – key frame systems – motion specification –morphing – tweening, Light Scources.	8 hours

Text Books				
Code	**Title of the Book**	**Author Name /Designation / Organization**	**Publisher**	**Edition/ Publication Year**
T1	Computer Graphics with OpenGL	Donald Hearn and Pauline Baker	Pearson	4th Edition, 2020
T2	Introduction to Computer Graphics and the Vulkan API	Kenwright	Cambridge Scholars Publishing	2021

Reference Books				
Code	**Title of the Book**	**Author Name/Designation/ Organization**	**Publisher**	**Edition/ Publication Year**
R1	Computer Graphics: Principles and Practice	John F. Hughes, Andries van Dam, Morgan McGuire, David F. Sklar, James D. Foley, Steven K. Feiner, and Kurt Akeley	Pearson	4th Edition, 2022
R2	Interactive Computer Graphics: A Top-Down Approach with WebGL	Edward Angel and Dave Shreiner	Pearson	8th Edition, 2022

Useful Links	
1	http://www.coursera.org/articles/computer-graphics
2	http://www.javatpoint.com/computer-graphics-tutorial

Semester	Course Code	Name of the Course	L	T	P	Credits
II	MC2L006	Data Base Management System LAB	-	-	4	2

Marks Distribution			
CA	**MSE**	**ESE**	**Total**
60	-	40	100

Course Outcomes		
Sr No	**Code**	**CO statement**
1	CO1	Understand and Execute Database Installation
2	CO2	Design and Implement Entity-Relationship Models
3	CO3	Perform Basic SQL Operations and Constraints
4	CO4	Manage Database Schema and Transactions
5	CO5	Write and Optimize SQL and PL/SQL Queries

List Of Experiments	
Sr No	**Description**
1	Introduction SQL and Oracle Installation.
2	Draw E-R diagram and convert entities and relationships to relation table for a given scenario. a. Two assignments shall be carried out i.e. consider two different scenarios (eg. bank, college).
3	Perform the following::- Data constraints (Primary key, Foreign key, Not Null) Data insertion into a table.
4	Perform the following: a. Viewing all databases, Creating a Database, Viewing all Tables in a Database, Creating Tables (With and Without Constraints), Inserting/Updating/Deleting Records in a Table, Saving (Commit) and Undoing (rollback).
5	Perform the following: a. Altering a Table, Dropping/Truncating/Renaming Tables, Backing up / Restoring a Database.
6	Implementation of different types of operators in SQL
7	For a given set of relation schemes, create tables and perform the following Simple Queries, Simple Queries with Aggregate functions, Queries with Aggregate functions (group by and having clause).
8	For a given set of relation schemes, create tables and perform the following Join QueriesInner Join, Outer Join Subqueries- With IN clause, With EXISTS clause.
9	Write a Pl/SQL program using FOR loop to insert ten rows into a database table
10	Writing SQL and PL/SQL queries to retrieve information from the databases.

Content Beyond Syllabus	
Sr No	**Description**
1	CASESTUDY (Student Progress Monitoring System or any other)
2	Study and Implementation of triggers

Semester	Course Code	Name of the Course	L	T	P	Credits
II	MC2L007	Data Science Using Python LAB	-	-	4	2

Marks Distribution			
CA	**MSE**	**ESE**	**Total**
60	-	40	100

Course Outcomes		
Sr No	**Code**	**CO statement**
1	CO1	Understand the fundamental concepts of data science and the role of Python in data processing and analysis.
2	CO2	Apply basic Python programming skills for data manipulation and script writing.
3	CO3	Perform data cleaning tasks, including handling missing data and outliers using Python.
4	CO4	Utilize Numpy and Pandas for efficient data processing and transformation.
5	CO5	Create visual representations of data using Matplotlib to aid in data analysis and interpretation.

List Of Experiments	
Sr No	**Description**
1	Set up Python using a virtual environment, create basic variables, and demonstrate different data types (int, float, string, list, tuple, dict, set).
2	Write a script that performs arithmetic operations, string manipulations (concatenation, slicing), and type conversions.
3	Import a dataset with missing values, inspect the data, and handle missing values using techniques like deletion, mean imputation, and forward/backward fill.
4	Use Python libraries like Pandas and Numpy to identify outliers using statistical methods (IQR, Z-score) and handle them by removing or capping.
5	Demonstrate normalization, standardization, and data encoding on a sample dataset using Python.
6	Create Numpy arrays, perform basic operations (addition, multiplication), and apply statistical functions (mean, median, standard deviation).
7	Load a dataset into a Pandas DataFrame, perform data filtering, grouping, and aggregation operations.
8	Create line and bar charts to represent trends in a dataset (e.g., time series data).
9	Use Matplotlib to create histograms for data distribution and scatter plots to identify relationships between variables.
10	Create pie charts to represent the proportion of different categories in a dataset.

Content Beyond Syllabus	
Sr No	**Description**
1	**Data Visualization with Seaborn**: Introduction to Seaborn for enhanced and more visually appealing data plots beyond basic Matplotlib functionalities.
2	**Time Series Data Analysis**: Introduction to time series analysis, including basic concepts and visualizations like trend lines and seasonal decomposition.

Semester	Course Code	Name of the Course	L	T	P	Credits
II	MC2L008	PHP LAB	-	-	4	2

Marks Distribution			
CA	**MSE**	**ESE**	**Total**
60	-	40	100

Course Outcomes		
Sr No	**Code**	**CO statement**
1	CO1	Understand the fundamentals of PHP and its usage in web development.
2	CO2	Learn how to connect PHP with databases and perform CRUD operations.
3	CO3	Develop secure PHP applications, ensuring proper input validation and session management.
4	CO4	Build and deploy dynamic web applications using PHP and MySQL.
5	CO5	Implement Object-Oriented Programming principles and advanced PHP concepts.

List Of Experiments	
Sr No	**Description**
1	Create a basic PHP script that prints "Hello World".
2	Write PHP scripts demonstrating the use of variables, data types, and constants.
3	Create custom PHP functions to calculate factorials, primes, etc.
4	Perform string operations (concatenation, slicing)
5	Create a PHP form to collect user details, validate inputs, and handle form submission securely.
6	Create a file upload system in PHP where users can upload images and files.
7	Implement session-based user login and logout functionality.
8	Implement Create, Read, Update, and Delete (CRUD) operations on a user management system.
9	Create a simple RESTful API in PHP to handle user information (GET, POST, PUT, DELETE requests).
10	Build a basic content management system (CMS) using PHP, MySQL, and OOP principles.

Content Beyond Syllabus	
Sr No	**Description**
1	Integrating third-party APIs in PHP projects (Google Maps, Payment Gateway).
2	Performance optimization techniques in PHP (caching, code optimization).